



juna.ai

Status Meeting 17.03.2026

DS Smith

# Project status

## Data Integration

- Live data connection with frequency 1/min

## Optimization

- 670 tags streamed into Centerlining
- Centerlining tool with feedback loop (<https://dssmith.juna.ai/optimization/>)

## Juna AI Agents

- Factory Model (<https://dssmith.juna.ai/factory/>)
- Knowledge base (<https://dssmith.juna.ai/settings/knowledge-base/>)
- Agents examples:
  - Yesterday Machine performance analysis ([Link to Chat](#))
  - 30 days sheet break root cause analysis with background agent ([Link to Chat](#))
  - Analysis of overall production efficiency and improvement actions ([Link to Chat](#))



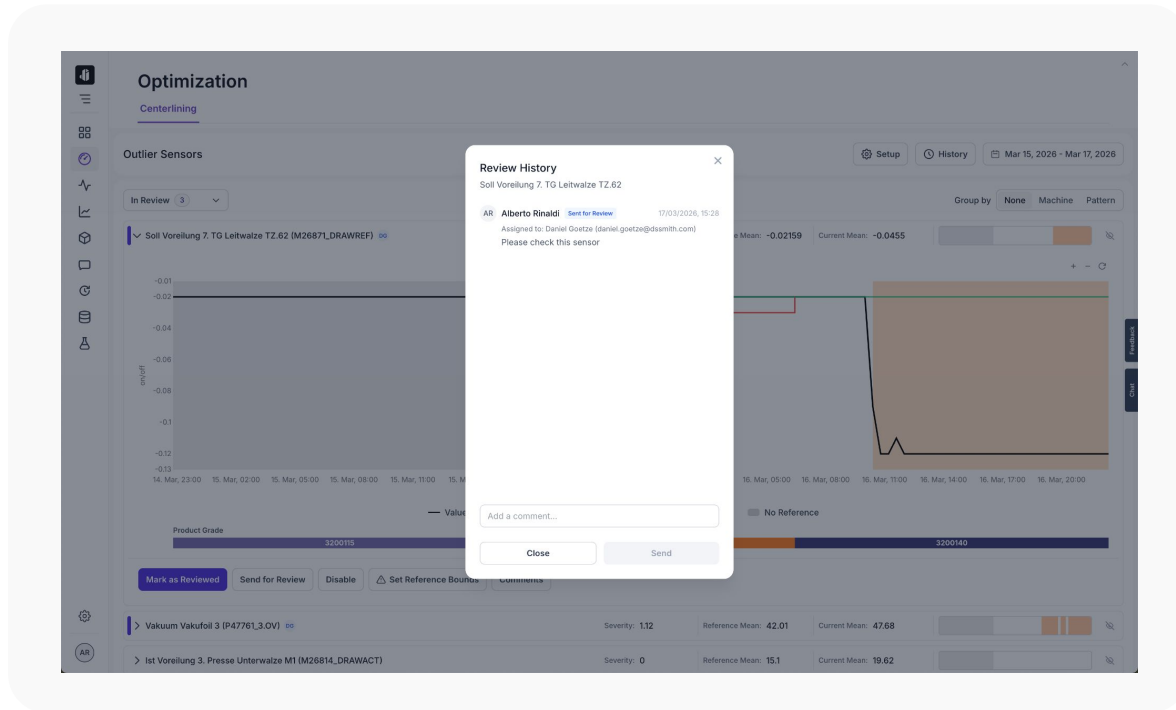
# Process monitoring

Outliers detection based on historical golden batch analysis



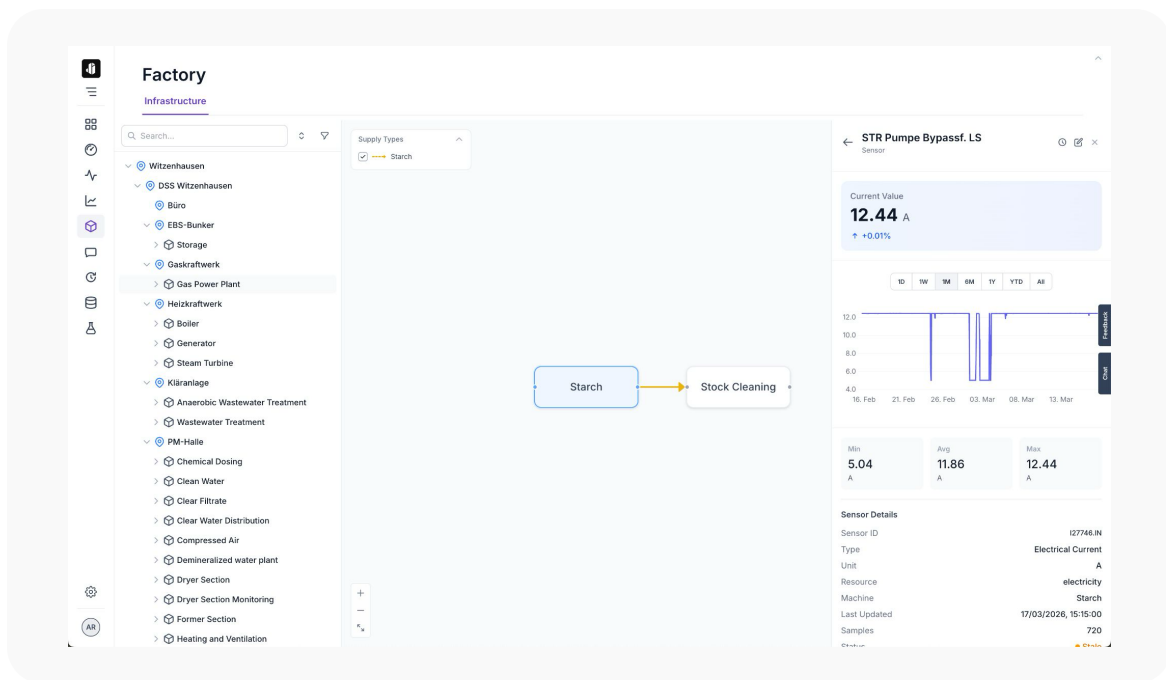
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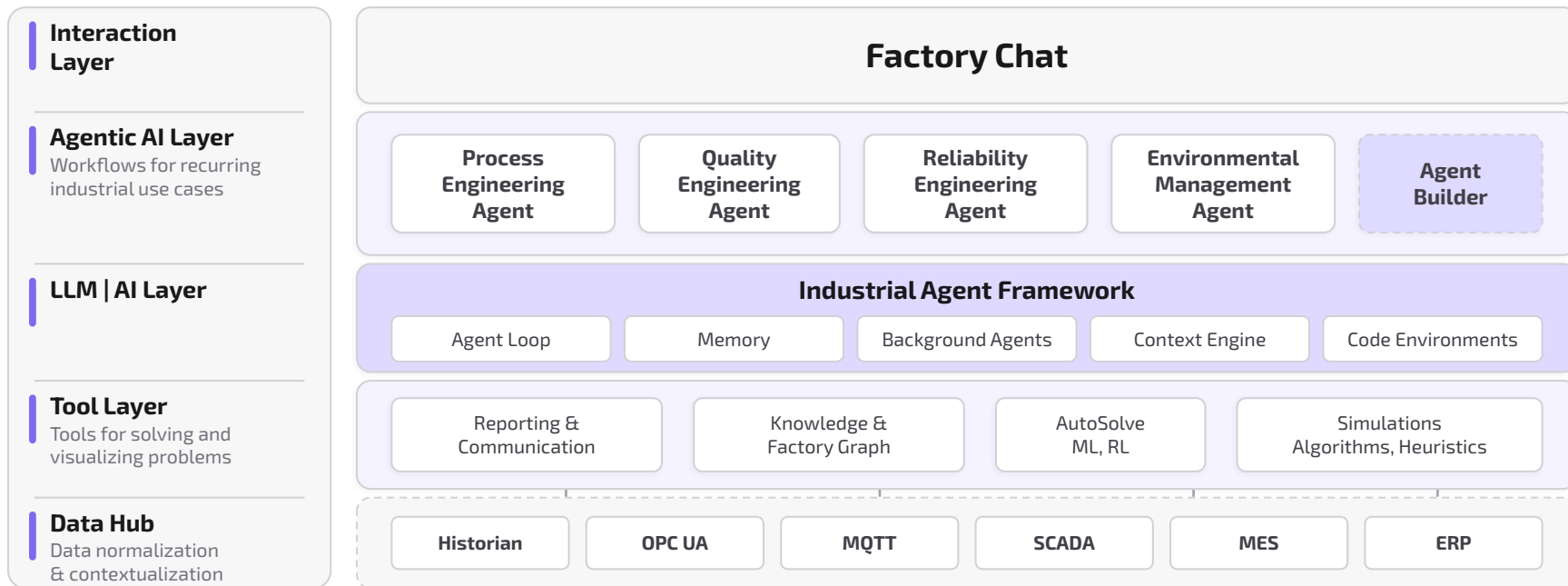
# Juna Factory Graph - The Foundation

A real-time map of resource, product and energy flows across your factory, which builds the foundation for Agentic optimization



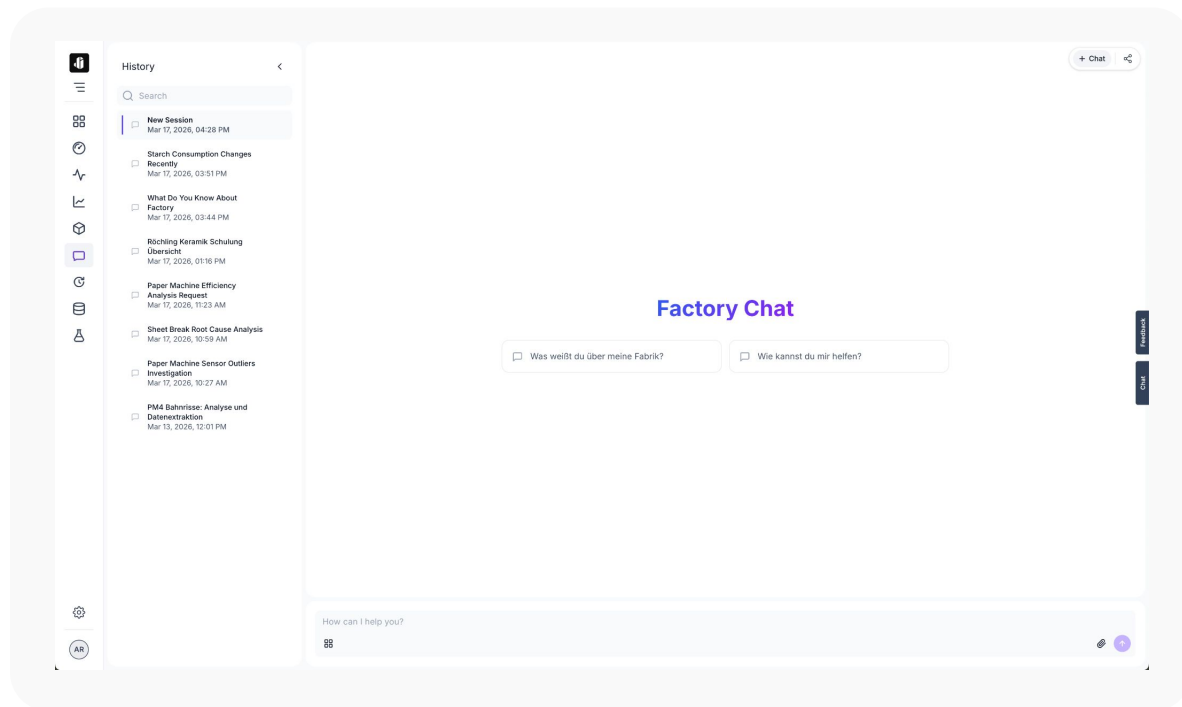
# Juna Agentic Factory OS

The leading AI-native decision making platform that orchestrates industrial AI Agents



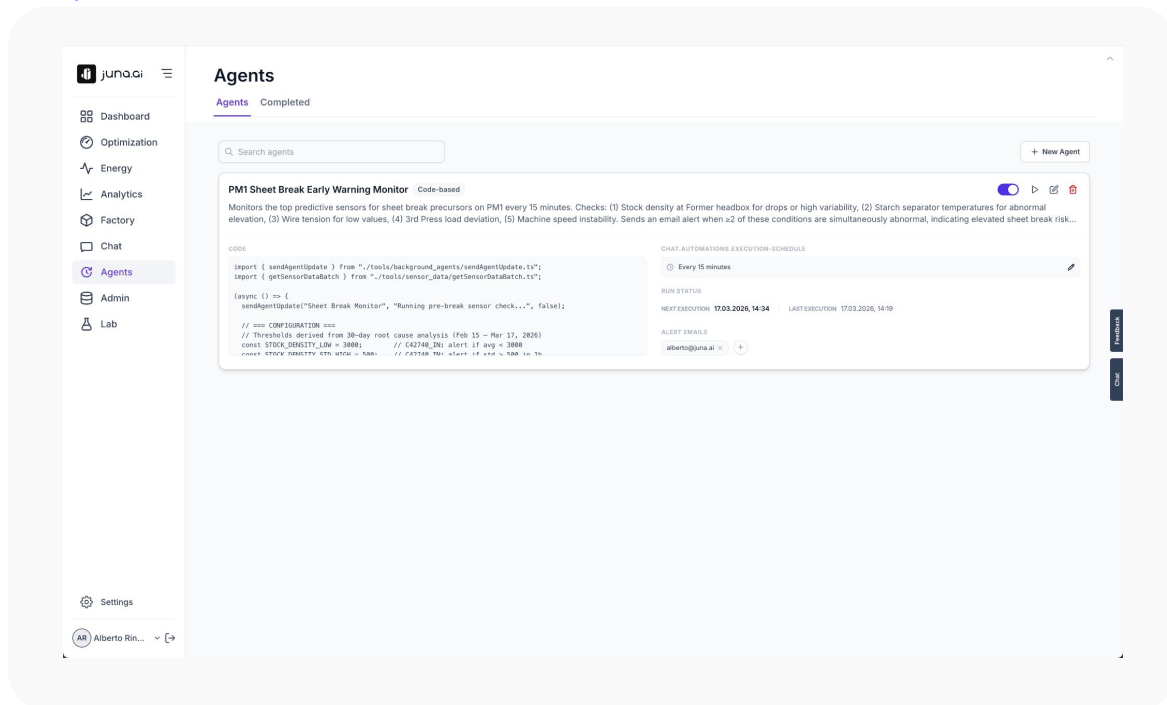
# The Factory Chat

Talk to your factory, perform custom data science analysis on real-time production data, build recurring background agents, all in natural language



# The Juna Agent Builder

Lets you build any industrial AI Agent you can think of in natural language, to automate analysis, reports or anomaly detection





# The Juna Agents in Action

AI Agents analyze, report, reason and continuously help you improve your factory



## Alert: Quality Monitor - Surface Area Deviation

Hello,

Your background agent **Quality Monitor - Surface Area Deviation** identified 1 new alerts.

### ALERT SUMMARY

- ⚠️ Surface Area Deviation Detected: BETBE out of specification on: Anlage 2 (81.8 m²/g), Anlage 5 (95.0 m²/g). Initiate PEBxyz procedure: Review feedstock quality, reactor temperature profiles, and quench timing. Consult Quality Engineering team.

Please check the details in your dashboard.

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## Alert: Weekly Energy Consumption Monitor

Hello,

Your background agent **Weekly Energy Consumption Monitor** identified 1 new alerts.

### ALERT SUMMARY

- ⚠️ Energy Alerts Detected: 6 alert(s) require attention:
  - Factory Total: Exceeded weekly threshold (414274 > 300000 kWh)
  - NH3 Plant: Exceeded weekly threshold (202226 > 160000 kWh)
  - Production Lines: Exceeded weekly threshold (111306 > 70000 kWh)
  - Mixplant: Exceeded weekly threshold (26810 > 20000 kWh)
  - Compressed Air: +19.1% increase (15836 vs 13298 kWh)
  - Compressed Air: Exceeded weekly threshold (15836 > 10000 kWh)

### \*\*Weekly Energy Summary\*\*

- \*\*Factory Total:\*\* 414274 kWh (+2.0% vs last week)
- \*\*NH3 Plant:\*\* 202226 kWh (48.8% of total) (-4.0% vs last week)

### \*\*Area Breakdown:\*\*

- NH3 Plant: 202226 kWh (-4.0%)
- Production Lines: 111306 kWh (+7.8%)
- Mixplant: 26810 kWh (+13.5%)
- Boilers: 9428 kWh (-0.6%)
- Compressed Air: 15836 kWh (+19.1%)

Please check the details in your dashboard.

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## Alert: Long-term Emission Trend Analysis

Hello,

Your background agent **Long-term Emission Trend Analysis** identified 1 new alerts.

### ALERT SUMMARY

- ⚠️ Emission Anomalies Detected: \*\*Monthly Emission Trend Analysis Complete\*\*

Analysis Period: 2025-02 to 2026-02  
Total Months Analyzed: 13

### \*\*Key Statistics:\*\*

- Average Monthly Emissions: 439.92 tonnes CO2
- Standard Deviation: 176.90 tonnes CO2
- Min Month: 73.83 tonnes CO2
- Max Month: 664.14 tonnes CO2

### \*\*Findings:\*\*

- ✓ Recent 3-month average (237.05 t CO2) is 46.1% below historical average - good progress!
- 🔍 Outlier months requiring investigation: 2026-02
- 📉 Months with >10% emission increase: 2025-03 (+51.1%), 2025-05 (+23.6%), 2025-07 (+11.7%), 2025-10 (+22.0%), 2025-11 (+44.5%)
- 📊 Emission source breakdown: Gas 21.7%, Electricity 78.3%

### \*\*Recommendations:\*\*

- Investigate root causes for outlier months (equipment issues, production anomalies, or data quality)
- Review operational changes during months with significant increases



# The Juna Chat


History

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New Session  
Mar 17, 2026, 04:28 PM

Starch Consumption Changes  
Recently  
Mar 17, 2026, 03:51 PM

What Do You Know About  
Factory  
Mar 17, 2026, 03:44 PM

Röchling Keramik Schulung  
Übersicht  
Mar 17, 2026, 01:16 PM

Paper Machine Efficiency  
Analysis Request  
Mar 17, 2026, 11:23 AM

Sheet Break Root Cause Analysis  
Mar 17, 2026, 10:59 AM

Paper Machine Sensor Outliers  
Investigation  
Mar 17, 2026, 10:27 AM

PM4 Bahnrisse: Analyse und  
Datenextraktion  
Mar 17, 2026, 12:01 PM

 **Sensors to Monitor for Early Warning**

Based on the correlation analysis, these sensors showed the strongest pre-break signals:

| Priority | Sensor                       | Ref ID                | What to Watch                      | Alert Threshold                 |
|----------|------------------------------|-----------------------|------------------------------------|---------------------------------|
| 1        | Stock Density Headbox Former | C42740_IN             | Rapid drops or high variability    | <3,000 or std >500 in 1h window |
| 2        | Starch Separator Temp Former | T27760.0V             | Sustained elevation                | >30°C                           |
| 3        | Wire Tension 7TGO            | K19770_1.IN           | Low tension                        | <3.0 kN                         |
| 4        | 3rd Press Section 1          | P46631A.CON           | Load deviation                     | >77                             |
| 5        | Starch Separator Temp LS     | T27750.0V             | Elevated temperature               | Abnormal rise                   |
| 6        | Exhaust Fan 3 Current        | I25803.IN             | Drops (dryer instability)          | Sudden decrease                 |
| 7        | Master Reel Speed            | S26791.IN             | Speed variability >100 m/min range | Speed range >200 in 1h          |
| 8        | HD Spray Former/SAW          | P42779.PV , P42797.PV | Pressure deviations                | Outside centerlining bounds     |

 **Recommended Preventive Measures**

Immediate Actions

- Investigate the starch system** — Separator temperatures (LS and Former) are running dramatically outside reference bounds. This is likely a root issue affecting sheet bonding strength. Check for fouling, valve settings, or cooling system faults.
- Review wire tension control** — 6 breaks in one 7-day period all had low wire tension. Check tension controller calibration and mechanical condition of tensioning systems.
- Stabilize stock density feed** — The high variability in headbox stock density is the most common precursor. Review stock preparation consistency control loops and mixing pump behavior.

Operator Monitoring Protocol

- Pre-break alert window:** Monitor the Priority 1-3 sensors above. When ≥2 of these 3 sensors are simultaneously abnormal, the risk of a sheet break within the next 1-2 hours is significantly elevated.
- Post-stop ramp-up:** After extended stops, allow extra time for stabilization before pushing to full speed. The 3 post-restart breaks suggest operators may be ramping too fast.
- Afternoon shift awareness:** 40% of breaks occur between 12:00-18:00 CET — this may correlate with grade changes or shift transitions.

Centerlining Corrections

How can I help you?



Feedback

Chat

# The Juna Chat



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+ Chat



## Recommendations for Operators

**1. Reduce speed proactively BEFORE or AT the grade change — not hours later**

> (Derived) In both Feb 14 and Mar 16, the machine ran too fast for 3200140 initially. The reference data shows 3200140 has a target speed of ~1013 m/min (reference mean). When coming from 3300130 at >1030 m/min, **step the speed down to target at the point of grade change**, don't wait for instability to force it.

**2. Watch headbox consistency (C42740\_IN) closely in the first 2-6 hours**

> (Derived) Consistency instability (std >100) in the post-GC window is a strong leading indicator of breaks. On Feb 14 and Mar 16, the consistency was highly variable. On clean transitions (Feb 06, Mar 10), it stayed very tight (std <25). Consider tightening consistency control loop tuning during grade transitions, or temporarily narrowing consistency alarm limits.

**3. Pre-adjust starch separator temperature**

> (Observed) The reference bounds show a massive operating point shift for the starch separator (Former): 3300130 runs at ~15.9°C mean vs. 3200140 at ~0.2°C. This suggests starch to the former may be largely inactive on 3200140. Operators should switch the starch configuration at the time of grade change, not let it drift.

**4. Pre-configure steam pressures for the target grade**

> (Observed) Steam cylinder pressures (groups 19-27, 28-35, 36-41) showed up as newly outlier sensors during yesterday's events. The reference bounds for 3200140 are much tighter ( $\sigma = 0.02$  bar) compared to 3300130 ( $\sigma = 0.3-0.4$  bar). Pre-loading the dryer steam profile for 3200140 at grade change time would reduce the settling period.

**5. Consider a grade-change checklist for 3300130 → 3200140**

> (Assumed — no existing procedure found in knowledge base) The knowledge base does not contain a documented grade change procedure (Sortenwechsel) for this transition. Given that this is the **most frequent grade change** (6x in 90 days) and has a **50% break rate**, it would be worthwhile to create a standardized checklist:

- > - Target speed for 3200140
- > - Starch separator on/off for Former
- > - Steam profile presets per cylinder group
- > - Consistency setpoint and alarm limits
- > - Drive draw/load-sharing references

**6. Review optimization reference bounds for 3200140**

> (Observed) The 3200140 bounds for several key sensors show extremely low sigma values (e.g., press air pressure  $\sigma=0.06$ , steam pressure  $\sigma=0.04$ ), meaning the bounds were likely trained on a very narrow operating window. This makes even minor deviations trigger outliers. Consider whether these bounds need retraining with more representative data, or whether the tight bounds are intentional.

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**Bottom line:** The strongest lever is **proactive speed reduction at grade change time**. The data clearly shows that when operators wait hours to adjust speed, consistency destabilizes and breaks follow. A documented recipe for 3300130 → 3200140 covering speed, starch, and steam presets would eliminate much of the current 50% break rate.

Not useful 1 2 3 4 5 Very useful

How can I help you?

